CHAPTER 2

# Reading on Paper and on Screens: The State of the Evidence

The question this thesis takes up is older than the screens it is usually asked about. Long before the first e-reader, typographers measured how the size of a letter, the length of a line and the space between lines changed the speed and the comfort of reading (Tinker, 1963). What changed with screens was the scale of the experiment: by the 2010s a large share of what students read for their courses reached them through a display, and the question of whether that mattered left the laboratory for the lecture hall (Baron, 2015).

This chapter reviews what is known. **Section 2.1** gathers the studies of comprehension, **Section 2.2** those of speed and line length, and **Section 2.3** the arguments about attention that frame the rest of the thesis. The method of the study itself follows in Chapter 3.

## 2.1 Comprehension

The most cited early comparison asked Norwegian upper-secondary students to read two texts, one narrative and one expository, either on paper or as PDF files on a computer screen (Mangen et al., 2013). Those who read on paper scored better on the comprehension test that followed. The difference was modest, but it pointed in the same direction as a growing number of studies, and it raised a question the authors could not settle: whether the advantage came from the medium, from the way readers navigated it, or from what they expected to do with it.

Two meta-analyses later pooled the evidence. Delgado et al. (2018) combined 54 studies with more than 170,000 participants and found an advantage for paper in the comprehension of informational texts, larger when reading was done under time pressure and, unexpectedly, larger in the more recent studies than in the older ones. Clinton (2019) reached a similar conclusion from a narrower set of experiments: reading from paper led to better comprehension, while the time spent reading did not differ between the media. Neither review found a reliable

difference for narrative texts. A pilot study run for this thesis with forty first-year students found the same pattern (Ibáñez Roca, 2023, pp. 12–15), although its sample was too small to settle it.

## 2.2 Speed and line length

Speed is the measure readers notice first and researchers trust least. A reader can go faster by understanding less, and a gain in words per minute says nothing of what was retained (Rayner et al., 2016). Studies that control for comprehension tell a more modest story than the claims of speed-reading courses: the eyes move in short jumps, take in a few letters on either side of the fixation, and cannot be trained to take in a whole line at once (Rayner et al., 2016).

Line length is the variable typographers have argued about longest. On screen, Dyson & Haselgrove (2001) measured speed and comprehension at several line lengths and found that the two do not always improve together, which warns against judging a layout by speed alone. Older work on print had already shown that the best length depends on the size of the type and the leading, not on a fixed count of characters (see also Ibáñez Roca, 2023, Section 2; Tinker, 1963).

## 2.3 Attention and the reading brain

The empirical studies leave room for a wider argument. Wolf (2018) holds that the habits formed by reading on screens, quick and skimming, carry over to the reading of long texts and weaken the slower processes that comprehension of a demanding argument needs. Baron (2015) reaches a similar view from surveys of students, many of whom reported that they concentrated better on paper even when they preferred screens for convenience. These are arguments rather than measurements, and this thesis treats them as hypotheses: Chapter 3 tests whether the advantage for paper reported in **Section 2.1** holds when the screen layout is set with the care a printed page receives.

## References

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